Cells, biological molecules and movement of substances

25 questions: 8 multiple choice, 17 structured.

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1 · Part 1

Paper 1 & 2 · Level of organisation · [1]

Which list shows the levels of organisation in an organism in order, from the smallest to the largest?

A cell → organelle → tissue → organ → system

B organelle → cell → tissue → organ → system

C organelle → tissue → cell → organ → system

D organelle → cell → organ → tissue → system

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2 · Part 1

Paper 1 & 2 · Cell structure · [1]

Which row correctly matches a cell structure to its function?

	structure	function
A	cell wall	contains cell sap
B	ribosome	stores the genetic information of the cell
C	vacuole	absorbs light energy for photosynthesis
D	cell membrane	controls which substances enter and leave the cell

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3 · Part 1 Paper 2 only Cell structure [1]

Doctors are considering treating a patient with stem cells taken from an embryo.

Which statement is a disadvantage of this treatment?

A The stem cells may divide in an uncontrolled way and form a tumour.

B The stem cells can differentiate into many different types of specialised cell.

C The stem cells can replace cells that have been damaged by disease.

D The stem cells can be grown in large numbers in a laboratory.

4 · Part 1 Paper 1 & 2 Biological molecules [1]

Which chemical element is present in all proteins but is not present in lipids?

A carbon

B hydrogen

C nitrogen

D oxygen

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5 · Part 1 Paper 1 & 2 Biological molecules [1]

Mei tested a liquid food. The table shows her results.

test	result
Benedict's solution added and the mixture heated in a water bath	colour changed from blue to brick red
iodine solution added	colour stayed orange-brown
biuret solution added	colour changed from blue to purple

Which substances does the food contain?

A glucose and starch

B glucose and protein

C starch and protein

D glucose, starch and protein

6 · Part 1 Paper 1 & 2 Biological molecules [1]

An enzyme-controlled reaction is carried out at 5 °C. The rate of reaction is very slow.

Why is the rate of reaction slow at 5 °C?

A The enzyme is killed by the low temperature.

B The enzyme is used up quickly in the reaction.

C The substrate molecules have changed shape.

D The enzyme and substrate molecules have less kinetic energy, so they collide less often.

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7 · Part 1

Paper 2 only | Biological molecules | [1]

In an investigation into the effect of pH on the activity of an enzyme, a buffer solution is added to each reaction mixture.

Why is a buffer solution added?

A to provide the substrate for the enzyme

B to speed up the reaction

C to keep the pH of each mixture at a fixed value

D to provide energy for the reaction

8 · Part 1

Paper 1 & 2 | Movement of substances into and out of cells | [1]

Cells lining the small intestine absorb glucose. They continue to absorb glucose when the concentration of glucose in the small intestine is lower than the concentration inside the cells.

Which process is used, and why?

A active transport, because glucose moves against its concentration gradient using energy from respiration

B diffusion, because glucose moves down its concentration gradient

C osmosis, because glucose moves through a partially permeable membrane

D diffusion, because glucose moves using energy from respiration

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9 · Part 1

Paper 1 & 2 Level of organisation [5]

The table lists five biological structures.

structure	level of organisation
chloroplast	
red blood cell	
xylem	
leaf	
digestive system	

(a) Complete the table by writing organelle, cell, tissue, organ or system next to each structure. Use each word once. [2]

(b) State what is meant by a tissue. [1]

(c) The wall of the stomach contains muscle tissue, glandular tissue and epithelial tissue.

Explain why the stomach is described as an organ. [2]

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10 · Part 1

Paper 1 & 2 Cell structure [6]

Lucia used a light microscope to look at a cell from the leaf of a plant and a cell from the lining of her cheek.

(a) Complete the table by writing yes or no in each empty box to show whether each structure is present in each cell.

structure	leaf cell	cheek cell
cell wall		
chloroplasts		
nucleus		
large vacuole containing cell sap		

[2]

(b) Describe the function of the nucleus. [2]

(c) Cells from the roots of the same plant do not contain chloroplasts.

Suggest why root cells do not contain chloroplasts. [2]

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11 · Part 1

Paper 1 & 2 Cell structure [6]

The table shows the approximate mean number of mitochondria in three types of human cell.

type of cell	approximate mean number of mitochondria per cell
heart muscle cell	5000
liver cell	2000
skin cell	250

(a) Calculate how many times more mitochondria there are in a heart muscle cell than in a skin cell. [1]

(b) Explain why heart muscle cells contain so many mitochondria. [3]

(c) Some cells in the pancreas make and release large amounts of enzymes. These enzymes are proteins.

Suggest which cell structure, other than mitochondria, is present in very large numbers in these cells. Give a reason for your answer. [2]

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12 · Part 1

Paper 2 only

Cell structure

[6]

After fertilisation, a human zygote divides by mitosis many times to form an embryo. At first, all the cells in the embryo are similar. As the embryo develops, the cells differentiate.

(a) State what is meant by cell differentiation. [1]

(b) Explain why cell differentiation is important in the development of a multicellular organism. [3]

(c) In flowering plants, small groups of cells at the tips of roots and shoots stay unspecialised for the whole life of the plant. These cells keep dividing.

Suggest why this is an advantage to the plant. [2]

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13 · Part 1

Paper 2 only

Cell structure

[7]

Stem cells can be used to treat some diseases. The table gives information about two sources of stem cells.

feature	embryonic stem cells	adult stem cells from bone marrow
where the cells come from	embryos a few days old, left over from fertility treatment	the bone marrow of the patient being treated
types of specialised cell the stem cells can become	almost any type of cell	a limited number of types, mainly blood cells
other information	the embryo is destroyed when the cells are removed	a needle must be pushed into one of the patient's bones to remove the cells

(a) Use the information in the table to explain why embryonic stem cells could be used to treat more diseases than adult stem cells from bone marrow. [2]

(b) Discuss the advantages and disadvantages of treating a patient with embryonic stem cells rather than with stem cells from the patient's own bone marrow. [5]

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14 · Part 1

Paper 2 only

Cell structure

[8]

During a heart attack, some of the muscle cells in the heart can die. Doctors carried out a trial with twenty patients whose heart muscle had been damaged by a heart attack. The patients were split into two groups of ten.

Patients in group 1 were injected in the heart with stem cells. Patients in group 2 were injected in the heart with a salt solution that contained no cells.

The doctors measured the percentage of the blood in the heart that was pumped out with each heartbeat. They measured this before the injection and again six months after the injection.

group	mean value before injection / %	mean value six months after injection / %
1 (stem cells)	38	45
2 (salt solution)	39	41

(a) Calculate the percentage increase in the mean value for group 1.

Use the formula: percentage increase = (value after − value before) ÷ value before × 100 [2]

(b) Explain why the doctors included group 2 in the trial. [2]

(c) Suggest how the stem cells could have improved the pumping of the heart. [2]

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(d) A newspaper reported that "stem cells cure heart damage".

Evaluate this claim using the information given. [2]

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15 · Part 1 Paper 1 & 2 Biological molecules [5]

Starch, glycogen and protein are large molecules made from smaller basic units.

(a) Complete the table to show the smaller basic units that each large molecule is made from.

large molecule	smaller basic units
starch	
glycogen	
protein	

[2]

(b) State the chemical elements present in starch. [1]

(c) Liver cells store glucose as glycogen. Glucose is soluble in water. Glycogen is insoluble in water.

Suggest why it is an advantage for liver cells to store glycogen rather than glucose. [2]

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16 · Part 1

Paper 1 & 2 Biological molecules [6]

Jamal wants to find out whether peanuts contain fat. He crushes some peanuts and uses ethanol to test them.

(a) Describe how Jamal should test the crushed peanuts for fat, and the result he would see if fat is present. [3]

(b) Jamal also carries out the same test using water instead of crushed peanuts.

Explain why Jamal carries out this test with water. [2]

(c) Ethanol is flammable.

State one safety precaution Jamal should take when he uses ethanol. [1]

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17 · Part 1

Paper 1 & 2 | Biological molecules | [5]

Biological washing powders contain enzymes. Two of these enzymes are a protease and a lipase.

(a) Enzymes are biological catalysts. State what is meant by a catalyst. [1]

(b) The lipase helps to remove greasy stains by digesting lipids.

State the two products formed when lipids are digested. [2]

(c) Explain why the protease in the washing powder does not digest lipids. [2]

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18 · Part 1

Paper 1 & 2 | Biological molecules | [6]

Some bacteria live in hot springs, where the water temperature is about 70 °C. Scientists compared the activity of an enzyme from these bacteria with the activity of the same type of enzyme from humans. The table shows their results.

temperature / °C	rate of reaction with human enzyme / arbitrary units	rate of reaction with hot-spring enzyme / arbitrary units
20	12	4
30	25	8
40	38	15
50	20	26
60	2	40
70	0	52
80	0	30

(a) Describe how the rate of reaction with the hot-spring enzyme changes as the temperature increases from 20 °C to 80 °C. [2]

(b) Identify the temperature in the table at which each enzyme had its highest rate of reaction. [1]

(c) Explain why the rate of reaction with the human enzyme is 0 at 70 °C. [3]

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19 · Part 1

Paper 1 & 2 | Biological molecules | [7]

Milk powder mixed with water forms a cloudy suspension because it contains protein. When a protease digests the protein, the suspension becomes clear.

Hana investigated the effect of temperature on the activity of a protease. This is her method.

1. Put 5 cm^3 of milk suspension into one test tube and 1 cm^3 of protease solution into another test tube.
2. Stand both tubes in a water bath at 20 °C for 5 minutes.
3. Pour the protease solution into the milk suspension and start a stop clock.
4. Record the time taken until a black cross on a card behind the tube can be seen clearly through the mixture.
5. Repeat steps 1 to 4 at 30 °C, 40 °C, 50 °C and 60 °C.

Hana did the whole investigation three times. The table shows her results.

temperature / °C	**time 1 / s**	**time 2 / s**	**time 3 / s**	**mean time / s**
20	310	290	300	300
30	150	162	156	156
40	84	78	81	
50	62	95	64	
60	did not clear in 600 s	did not clear in 600 s	did not clear in 600 s	

(a) Explain why Hana stood both tubes in the water bath for 5 minutes before mixing them. [2]

(b) Calculate the mean time at 40 °C. [1]

(c) Identify the anomalous result in the table. Calculate the mean time at 50 °C without the anomalous result. [2]

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(d) Hana concludes that the optimum temperature of the protease is 50 °C.

Suggest how Hana could find the optimum temperature more precisely. [2]

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20 · Part 1

Paper 1 & 2 Biological molecules [6]

Enzyme X is a protease from the stomach. Enzyme Y is a protease from the small intestine. The table shows the activity of each enzyme at different pH values.

pH	activity of enzyme X / arbitrary units	activity of enzyme Y / arbitrary units
1	30	0
2	48	0
4	20	5
6	2	22
8	0	45
10	0	18

(a) Describe the effect of pH on the activity of enzyme X. [2]

(b) Explain why enzyme X has no activity at pH 8. [3]

(c) The contents of the stomach have a pH of about 2.

Use the table to suggest why the contents of the small intestine need to have a higher pH than the contents of the stomach. [1]

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21 · Part 1

Paper 2 only

Biological molecules

[7]

Catalase is an enzyme that breaks down hydrogen peroxide into water and oxygen.

Leo investigated the effect of pH on the activity of catalase. For each pH, he mixed 2 cm^3 of catalase solution, 5 cm^3 of a buffer solution and 5 cm^3 of hydrogen peroxide solution in a flask at 25 °C. He collected the oxygen given off in a gas syringe for 2 minutes. He did one test at each pH.

pH	volume of oxygen collected in 2 minutes / cm^3
4	5
5	12
6	22
7	30
8	26
9	10

(a) State the dependent variable in Leo's investigation. [1]

(b) Calculate the rate of oxygen production at pH 7.

Use the formula: rate = volume of oxygen collected ÷ time

Give your answer in cm^3 per minute. [2]

(c) Leo concludes that the optimum pH of catalase is exactly pH 7.

Evaluate this conclusion. [2]

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(d) Describe how Leo could show that the oxygen was produced because of the catalase, and not because the hydrogen peroxide breaks down on its own. [2]

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22 · Part 1

Paper 2 only

Biological molecules

[7]

Amylase is an enzyme that digests starch. Iodine solution can be used to show whether starch is still present in a sample.

Hamid has: amylase solution, starch solution, iodine solution, buffer solutions of pH 3, 5, 7, 9 and 11, a spotting tile, a water bath at 35 °C, test tubes, syringes, dropping pipettes and a stop clock.

(a) Design an investigation that Hamid could carry out to find out how pH affects the activity of amylase. [6]

(b) State one safety precaution Hamid should take. [1]

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23 · Part 1

Paper 1 & 2 Movement of substances into and out of cells [6]

Amoeba is a single-celled organism that lives in fresh water. It has a cell membrane but no cell wall. The cytoplasm of *Amoeba* is more concentrated than the pond water around it.

(a) Oxygen enters *Amoeba* by diffusion.

State what is meant by diffusion. [2]

(b) Water enters the *Amoeba* cell all the time.

Explain why water enters the cell. [2]

(c) *Amoeba* has a structure called a contractile vacuole. The contractile vacuole collects water from the cytoplasm and pumps it out of the cell.

Suggest what would happen to an *Amoeba* if its contractile vacuole stopped working. [2]

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24 · Part 1

Paper 1 & 2 Movement of substances into and out of cells [7]

Agar jelly was made containing an indicator that is pink. The indicator turns colourless when acid reaches it. Cubes of the pink agar with sides of 1 cm, 2 cm and 3 cm were cut and placed in dilute hydrochloric acid. The time taken for each cube to become completely colourless was recorded.

length of side / cm	surface area / cm^2	volume / cm^3	surface area : volume ratio	time to become colourless / min
1	6	1	6 : 1	4
2				16
3	54	27	2 : 1	36

For a cube: surface area = 6 × (length of side)^2 and volume = (length of side)^3

(a) Calculate the surface area : volume ratio of the 2 cm cube. Show your working. [2]

(b) Describe the relationship between the surface area : volume ratio and the time taken for a cube to become colourless. [2]

(c) Explain why the 3 cm cube took the longest time to become colourless. [2]

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(d) State one variable that should be kept the same for each cube. [1]

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25 · Part 1

Paper 1 & 2 Movement of substances into and out of cells [8]

Five cylinders of potato were blotted dry and weighed. Each cylinder was placed in a sucrose solution of a different concentration. After two hours, the cylinders were removed, blotted dry and weighed again.

concentration of sucrose solution / %	initial mass / g	final mass / g	percentage change in mass / %
0	2.40	2.76	+15.0
5	2.50	2.65	+6.0
10	2.45	2.40	
15	2.40	2.22	−7.5
20	2.50	2.20	−12.0

(a) Explain why the percentage change in mass was calculated, rather than just the change in mass. [2]

(b) Calculate the percentage change in mass of the cylinder in the 10 % sucrose solution.

Use the formula: percentage change = (final mass − initial mass) ÷ initial mass × 100 [2]

(c) Explain the change in mass of the cylinder in the 20 % sucrose solution. [3]

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(d) Estimate the concentration of sucrose solution in which a potato cylinder would not change in mass. [1]

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